# Effects of xenoestrogenic environmental pollutants on the proliferation of a human breast cancer cell line (MCF-7).

A human breast cancer cell line (MCF-7) was used to develop an in vitro screening assay for the detection of xenoestrogenic environmental pollutants. MCF-7 cells were cultured in DMEM containing 5% fetal bovine serum (FBS). An estrogenic response was defined as an increase in the frequency of proliferating MCF-7 cells, and was measured using a thymidine analog, bromodeoxyuridine, and flow cytometry. Di-2-ethylhexyl phthalate (DEHP) and 4-n-nonylphenol (4-n-NP) were used as model chemicals. The proliferation rate of S-phase cells after 24 h of exposure to various concentrations of 17beta-estradiol and to model compounds was compared with a positive and a negative control, containing 1 nM 17beta-estradiol and 0.1% ethanol, respectively. DEHP and 4-n-NP increased the frequency of proliferating MCF-7 cells in a dose-dependent manner. The lowest concentration that significantly increased the proliferation of MCF-7 cells was 10 microM for DEHP and 1 microM for 4-n-NP. The results showed that the assay is accurate and quick to perform. It may prove a valuable tool for screening potential estrogen-mimicking environmental pollutants.